

Introduction to Analysis Homework 4

March, 20, 2025

1. (12 pt) Suppose that f is non-negative and continuous on $[1, 2]$ and that $\int_1^2 x^k f(x) dx = 5 + k^2$, for $k = 0, 1, 2$. Prove the following.
 - (a) $\int_1^4 f(\sqrt{x}) dx = 12$.
 - (b) $\int_{\frac{\sqrt{2}}{2}}^1 f(\frac{1}{x^2}) dx \leq \frac{5}{2}$.
 - (c) $\int_0^1 x^2 f(x+1) dx = 2$.
2. (10 pt) Suppose that E is Jordan measurable in $\mathbb{R}^n$, and that $f : E \rightarrow [0, \infty)$ is integrable. If $\int_E f(x) dx = 0$, show that $\{x \in E : f(x) \neq 0\}$ is measure zero.
3. (8 pt) Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$. Prove that the graph of $y = f(x)$, $x \in [a, b]$ is Jordan measurable in $\mathbb{R}^2$.
4. (10 pt) Suppose that $E \subset \mathbb{R}^n$ is Jordan measurable and that $f : E \rightarrow \mathbb{R}$ is integrable. If $f(E) \subset H$, for some compact set H , and $\phi : H \rightarrow \mathbb{R}$ is continuous, prove that $\phi \circ f$ is integrable on E , try **NOT** to apply Theorem 35.4.
5. (10 pt) Find $\int_E \sin(x+y) \cos(2x-y) dV$, where $E \subset \mathbb{R}^2$ is the region bounded by $y = 2x - 1$, $y = 2x + 3$, $y = -x$, $y = -x + 1$.